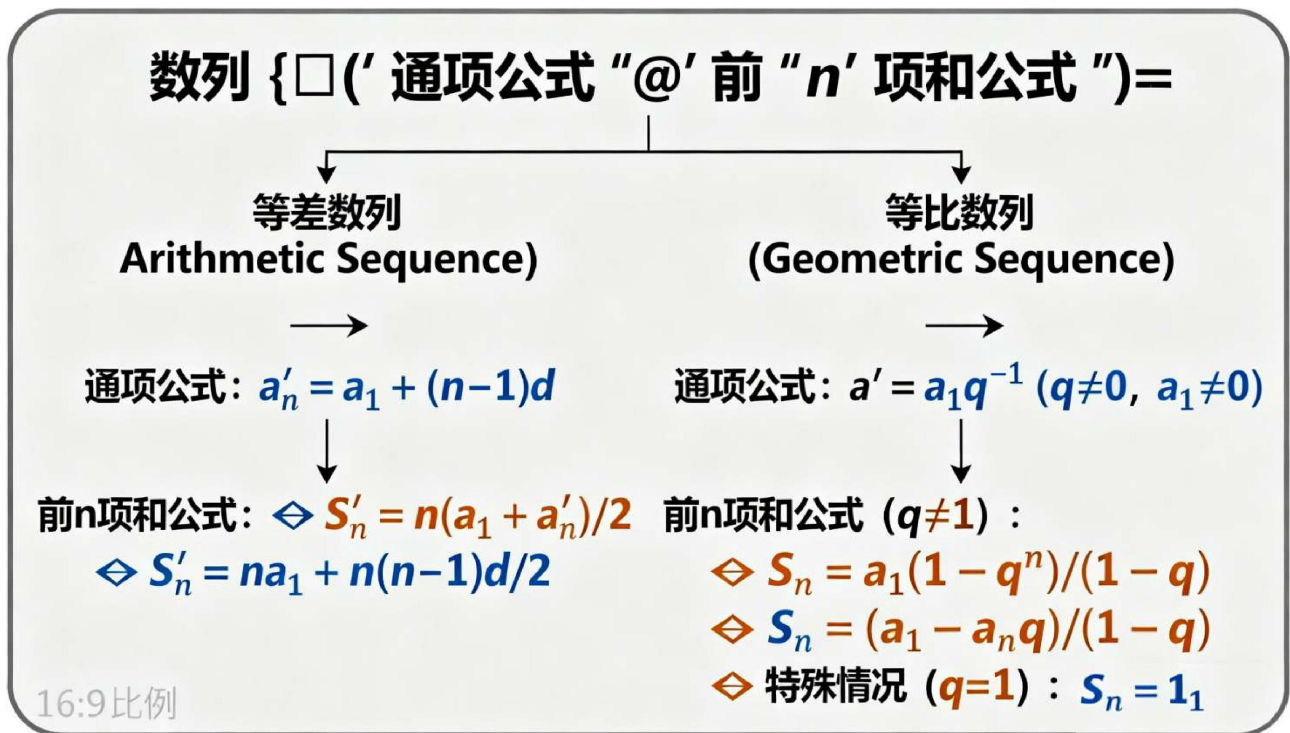


第六章 数列知识点详解

Chapter 7 Sequence Knowledge Detailed Explanation

一、本章知识点结构总结图

Chapter Knowledge Structure Diagram



二、数列基础(Sequence Basics)

2.1 核心定义(Core Definitions)

术语	完整定义(中文)	Complete Definition(English)
数列 Sequence	按照 一定顺序 排列的一系列数，记作 $\{a_n\}$ ，其中:- a_n 表示 “数列的第 n 项”(n 为正整数，称为 “项数”); - 例如:数列 $\{2,4,6,8\}$ 中， $a_1 = 2$ (首项), $a_2 = 4$ (第二项)。	A set of numbers arranged in a specific order , denoted as $\{a_n\}$, where:- a_n means “the n -th term of the sequence” (n is a positive integer, called “term number”);- Example: In the sequence $\{2,4,6,8\}$, $a_1 = 2$ (first term), $a_2 = 4$ (second term).
项 Term	数列中的 每个数 ，分为首项(a_1 , 第 1 项)、第 n 项(a_n)、末项(若有限数列，记为 a_m , m 为总项数)。	Each number in the sequence, including: - First term (a_1 , the 1st term); - n -th term (a_n); - Last term (for finite sequences, denoted as a_m , m = total number of terms).
通项公式 General Term Formula	表示 “数列第 n 项 a_n 与项数 n 之间对应关系” 的数学表达式，形如 $a_n = f(n)$ 。作用: 已知 n 可直接求 a_n ，或判断某数是否为数列中的项。	A mathematical expression that describes the relationship between the n -th term a_n and the term number n , in the form of $a_n = f(n)$. Function: Calculate a_n directly from n , or judge if a number is a term of the sequence.
前 n 项和 Sum of the First n Terms	数列前 n 项的总和，记作 S_n ，定义为 $S_n = a_1 + a_2 + a_3 + \dots + a_n$ 。作用:描述数列前 n 项的累积大小，是后续推导数列性质的基础。	The total sum of the first n terms of the sequence, denoted as S_n , defined as $S_n = a_1 + a_2 + a_3 + \dots + a_n$. Function: Describe the cumulative value of the first n terms, which is the basis for deriving sequence properties later.

2.2 关键区别:数列与集合(Key Distinction: Sequence vs Set)

对比维度	数列(Sequence)	集合(Set)
顺序性	有顺序(a_1, a_2, a_3 位置固定，交换后	无顺序($\{2,4,6\}$ 与 $\{6,4,2\}$ 是同

对比维度	数列(Sequence)	集合(Set)
	为不同数列)	一集合)
重复性	可重复(如数列{2,2,2}是合法的)	不可重复(集合中元素唯一, {2,2,2}等价于{2})
表示法	用 $\{a_n\}$ 表示(注意:不是集合符号, 仅表示数列)	用 $\{\}$ 表示(如 $\{x \mid x \text{ 是正偶数}\}$)

2.3 典型例题与解析(Typical Examples & Explanations)

例 1:由通项公式求特定项(Find Specific Term via General Term Formula)

题目:已知数列 $\{a_n\}$ 的通项公式为 $a_n = n + \frac{(n-1)^2}{2}$, 求 a_2 。

英文题目:Given the general term formula of sequence $\{a_n\}$ is $a_n = n + \frac{(n-1)^2}{2}$, find a_2 .

解析:

1. 明确目标:求“第 2 项”, 即 $n = 2$ 时的 a_n ;

2. 代入计算:将 $n = 2$ 代入通项公式:

$$a_2 = 2 + \frac{(2-1)^2}{2} = 2 + \frac{1}{2} = \frac{5}{2};$$

3. 结论: $a_2 = \frac{5}{2}$ (或 2.5)。

Explanation:

1. Clear goal: Find the “2nd term”, i.e., a_n when $n = 2$;

2. Substitute and calculate: Replace $n = 2$ into the general term formula:

$$a_2 = 2 + \frac{(2-1)^2}{2} = 2 + \frac{1}{2} = \frac{5}{2};$$

3. Conclusion: $a_2 = \frac{5}{2}$ (or 2.5).

例 2:由递推公式求特定项(Find Specific Term via Recursive Formula)

题目:已知数列 $\{a_n\}$ 满足 $a_n = \frac{a_{n-1}}{2^n+1} (n \geq 2)$, 且 $a_2 = 3$, 求 a_3 。

英文题目:Given sequence $\{a_n\}$ satisfies $a_n = \frac{a_{n-1}}{2^n+1} (n \geq 2)$ and $a_2 = 3$, find a_3 .

解析:

1. 递推公式含义:“第 n 项等于第 $n-1$ 项除以 (2^n+1) ”, 求 a_3 需先知道 a_2 ;
2. 代入计算: $n=3$ 时, $a_3 = \frac{a_2}{2^3+1} = \frac{3}{8+1} = \frac{3}{9} = \frac{1}{3}$;
3. 结论: $a_3 = \frac{1}{3}$ 。

Explanation:

1. Meaning of recursive formula: “The n -th term equals the $(n-1)$ -th term divided by (2^n+1) ”; to find a_3 , we first need a_2 ;
2. Substitute and calculate: When $n=3$, $a_3 = \frac{a_2}{2^3+1} = \frac{3}{8+1} = \frac{1}{3}$;
3. Conclusion: $a_3 = \frac{1}{3}$.

例 3:由前 n 项和求特定项(Find Specific Term via Sum of First n Terms)

题目:已知数列 $\{a_n\}$ 的前 n 项和公式为 $S_n = 3n - 4$, 求第 3 项 a_3 。

英文题目:Given the sum formula of the first n terms of sequence $\{a_n\}$ is $S_n = 3n - 4$, find the 3rd term a_3 .

关键公式:对任意数列, $a_n = \begin{cases} S_1 & (n=1) \\ S_n - S_{n-1} & (n \geq 2) \end{cases}$

(原因: $S_n = a_1 + a_2 + \cdots + a_{n-1} + a_n$, $S_{n-1} = a_1 + a_2 + \cdots + a_{n-1}$, 两式相减得 $a_n = S_n - S_{n-1}$, 但 $n=1$ 时 S_0 无意义, 故单独计算 $a_1 = S_1$)

解析:

1. 求 a_3 需用 $S_3 - S_2$ (因 $3 \geq 2$);
2. 计算 S_3 : $S_3 = 3 \times 3 - 4 = 9 - 4 = 5$;

3. 计算 S_2 : $S_2 = 3 \times 2 - 4 = 6 - 4 = 2$;

4. 求 a_3 : $a_3 = S_3 - S_2 = 5 - 2 = 3$;

5. 结论: $a_3 = 3$ 。

Explanation:

1. To find a_3 , use $S_3 - S_2$ (since $3 \geq 2$);

2. Calculate S_3 : $S_3 = 3 \times 3 - 4 = 9 - 4 = 5$;

3. Calculate S_2 : $S_2 = 3 \times 2 - 4 = 6 - 4 = 2$;

4. Find a_3 : $a_3 = S_3 - S_2 = 5 - 2 = 3$;

5. Conclusion: $a_3 = 3$.

三、等差数列(Arithmetic Sequence)

3.1 完整定义(Complete Definition)

若一个数列从**第二项**开始，每一项与它前一项的差都等于**同一个常数**，则这个数列称为等差数列。

数学表达式:对任意 $n \geq 2$ (n 为正整数), 有 $a_n - a_{n-1} = d$, 其中:

- d 称为“公差”(Common Difference), 可为正、负或零($d > 0$ 时数列递增, $d < 0$ 时递减, $d = 0$ 时为常数列, 如 $\{5, 5, 5\}$);
- 若 $n = 1$, 无“前一项”, 故首项 a_1 需单独给出(或通过其他条件推导)。

English Definition:

A sequence is called an arithmetic sequence if the difference between each term and its previous term is the **same constant** starting from the **second term**.

Mathematical expression: For any $n \geq 2$ (n is a positive integer), $a_n - a_{n-1} = d$, where:

- d is called “common difference”, which can be positive, negative, or zero (the sequence increases if $d > 0$, decreases if $d < 0$, and is a constant sequence if $d = 0$, e.g., $\{5, 5, 5\}$);
- For $n = 1$, there is no “previous term”, so the first term a_1 must be given separately (or derived from other conditions).

3.2 核心公式推导(Core Formula Derivation)

(1)通项公式(General Term Formula): $a_n = a_1 + (n - 1)d$

推导过程:

由等差数列定义, 有:

$$a_2 - a_1 = d \rightarrow a_2 = a_1 + d;$$

$$a_3 - a_2 = d \rightarrow a_3 = a_2 + d = a_1 + 2d;$$

$$a_4 - a_3 = d \rightarrow a_4 = a_3 + d = a_1 + 3d;$$

...

以此类推，第 n 项比首项多 $(n-1)$ 个公差，故：

$$a_n = a_1 + (n-1)d。$$

Derivation Process:

From the definition of arithmetic sequence:

$$a_2 - a_1 = d \rightarrow a_2 = a_1 + d;$$

$$a_3 - a_2 = d \rightarrow a_3 = a_2 + d = a_1 + 2d;$$

$$a_4 - a_3 = d \rightarrow a_4 = a_3 + d = a_1 + 3d;$$

...

By analogy, the n -th term is $(n-1)$ common differences more than the first term, so:

$$a_n = a_1 + (n-1)d。$$

(2)等差中项公式(Arithmetic Mean Formula):若 a, A, b 成等差数列，则 $A = \frac{a+b}{2}$

定义:若三个数 a, A, b 满足“ $A - a = b - A$ ”(即 A 与 a 的差等于 b 与 A 的差)，则 A 称为 a 与 b 的等差中项。

推导:

由 $A - a = b - A$ ，移项得 $2A = a + b$ ，故 $A = \frac{a+b}{2}$ 。

作用:判断三个数是否成等差数列(如:2, 5, 8 中， $A = \frac{2+8}{2} = 5$ ，故成等差)。

Definition:If three numbers a, A, b satisfy “ $A - a = b - A$ ” (i.e., the difference

between A and a equals the difference between b and A), then A is called the arithmetic mean of a and b .

Derivation:

From $A - a = b - A$, rearrange to get $2A = a + b$, so $A = \frac{a+b}{2}$.

Function: Judge if three numbers form an arithmetic sequence (e.g., in 2, 5, 8, $A = \frac{2+8}{2} = 5$, so they form an arithmetic sequence).

(3)前 n 项和公式(Sum of First n Terms Formula):两种形式

形式 1: $S_n = \frac{n(a_1+a_n)}{2}$ (已知首项 a_1 和末项 a_n 时用)

推导方法:倒序相加法(Reverse Order Addition):

设 $S_n = a_1 + a_2 + a_3 + \cdots + a_{n-1} + a_n$ ①;

将等式右边倒序, 得 $S_n = a_n + a_{n-1} + a_{n-2} + \cdots + a_2 + a_1$ ②;

①+②得: $2S_n = (a_1 + a_n) + (a_2 + a_{n-1}) + (a_3 + a_{n-2}) + \cdots + (a_n + a_1)$;

由等差数列性质, $a_1 + a_n = a_2 + a_{n-1} = a_3 + a_{n-2} = \cdots$ (共 n 组), 故:

$$2S_n = n(a_1 + a_n) \rightarrow S_n = \frac{n(a_1+a_n)}{2}.$$

Derivation Method: Reverse Order Addition:

Let $S_n = a_1 + a_2 + a_3 + \cdots + a_{n-1} + a_n$ ①;

Reverse the right side of the equation: $S_n = a_n + a_{n-1} + a_{n-2} + \cdots + a_2 + a_1$ ②;

Add ① and ②: $2S_n = (a_1 + a_n) + (a_2 + a_{n-1}) + (a_3 + a_{n-2}) + \cdots + (a_n + a_1)$;

From the property of arithmetic sequence, $a_1 + a_n = a_2 + a_{n-1} = a_3 + a_{n-2} = \cdots$ (total n groups), so:

$$2S_n = n(a_1 + a_n) \rightarrow S_n = \frac{n(a_1+a_n)}{2}.$$

形式 2: $S_n = na_1 + \frac{n(n-1)}{2}d$ (已知首项 a_1 和公差 d 时用)

推导:

将通项公式 $a_n = a_1 + (n-1)d$ 代入形式 1, 得:

$$S_n = \frac{n[a_1 + a_1 + (n-1)d]}{2} = \frac{n[2a_1 + (n-1)d]}{2} = na_1 + \frac{n(n-1)}{2}d。$$

Derivation:

Substitute the general term formula $a_n = a_1 + (n-1)d$ into Form 1:

$$S_n = \frac{n[a_1 + a_1 + (n-1)d]}{2} = \frac{n[2a_1 + (n-1)d]}{2} = na_1 + \frac{n(n-1)}{2}d。$$

3.3 典型例题与解析(Typical Examples & Explanations)

例 1:求等差数列的特定项(Find Specific Term of Arithmetic Sequence)

题目:在等差数列 $\{a_n\}$ 中, $a_1 = 8$, $d = -3$, 求 a_7 。

英文题目:In the arithmetic sequence $\{a_n\}$, $a_1 = 8$, $d = -3$, find a_7 .

解析:

1. 确定公式:已知 a_1 和 d , 用通项公式 $a_n = a_1 + (n-1)d$;

2. 代入参数: $n = 7$, $a_1 = 8$, $d = -3$, 得:

$$a_7 = 8 + (7-1) \times (-3) = 8 + 6 \times (-3) = 8 - 18 = -10;$$

3. 结论: $a_7 = -10$ 。

Explanation:

1. Determine the formula: Given a_1 and d , use the general term formula

$$a_n = a_1 + (n-1)d;$$

2. Substitute parameters: $n = 7$, $a_1 = 8$, $d = -3$, so:

$$a_7 = 8 + (7-1) \times (-3) = 8 + 6 \times (-3) = 8 - 18 = -10;$$

3. Conclusion: $a_7 = -10$ 。

例 2:已知两项求首项和公差(Find a_1 and d Given Two Terms)

题目:在等差数列 $\{a_n\}$ 中, $a_5 = 10$, $a_{12} = 31$, 求:(1) a_1 和 d ; (2) S_5 。

英文题目:In the arithmetic sequence $\{a_n\}$, $a_5 = 10$, $a_{12} = 31$, find: (1) a_1 and d ; (2) S_5 .

解析(1)求 和 :

1. 列方程:由通项公式, $a_n = a_1 + (n - 1)d$, 得:

- $a_5 = a_1 + (5 - 1)d = a_1 + 4d = 10$ ①;

- $a_{12} = a_1 + (12 - 1)d = a_1 + 11d = 31$ ②;

1. 解方程组:

方法一:减法消元(推荐, 更简便):

② - ①得: $(a_1 + 11d) - (a_1 + 4d) = 31 - 10 \rightarrow 7d = 21 \rightarrow d = 3$;

将 $d = 3$ 代入①: $a_1 + 4 \times 3 = 10 \rightarrow a_1 = 10 - 12 = -2$ 。

方法二:代入消元:

由①得: $a_1 = 10 - 4d$ ③;

将③代入②: $10 - 4d + 11d = 31 \rightarrow 10 + 7d = 31 \rightarrow 7d = 21 \rightarrow d = 3$;

再代入③: $a_1 = 10 - 4 \times 3 = -2$ 。

2. 结论: $a_1 = -2$, $d = 3$ 。

解析(2)求 :

1. 选择公式:已知 $a_1 = -2$, $a_5 = 10$, 用前 n 项和公式 $S_n = \frac{n(a_1 + a_n)}{2}$;

2. 代入计算: $n = 5$, 得:

$$S_5 = \frac{5 \times (-2 + 10)}{2} = \frac{5 \times 8}{2} = 20;$$

3. 结论: $S_5 = 20$ 。

Explanation (1) Find a_1 and d :

1. List equations: From the general term formula $a_n = a_1 + (n - 1)d$:

- $a_5 = a_1 + (5 - 1)d = a_1 + 4d = 10$ ①;
- $a_{12} = a_1 + (12 - 1)d = a_1 + 11d = 31$ ②;

1. Solve the system of equations:

Method 1: Elimination by Subtraction (Recommended, simpler):

$$\text{②} - \text{①}: (a_1 + 11d) - (a_1 + 4d) = 31 - 10 \rightarrow 7d = 21 \rightarrow d = 3;$$

Substitute $d = 3$ into ①: $a_1 + 4 \times 3 = 10 \rightarrow a_1 = 10 - 12 = -2$ 。

Method 2: Elimination by Substitution:

$$\text{From ①: } a_1 = 10 - 4d \text{ ③};$$

$$\text{Substitute ③ into ②: } 10 - 4d + 11d = 31 \rightarrow 10 + 7d = 31 \rightarrow 7d = 21 \rightarrow d = 3;$$

$$\text{Substitute into ③ again: } a_1 = 10 - 4 \times 3 = -2。$$

2. Conclusion: $a_1 = -2$, $d = 3$ 。

Explanation (2) Find S_5 :

1. Choose the formula: Given $a_1 = -2$, $a_5 = 10$, use the sum formula $S_n =$

$$\frac{n(a_1 + a_n)}{2},$$

2. Substitute and calculate: $n = 5$, so:

$$S_5 = \frac{5 \times (-2 + 10)}{2} = \frac{5 \times 8}{2} = 20;$$

3. Conclusion: $S_5 = 20$ 。

3.4 常见易错点(Common Pitfalls)

1. 忽略 “ $n \geq 2$ ” 的条件: 用 $a_n = S_n - S_{n-1}$ 求通项时, 需单独验证 $n = 1$ (如 $S_n =$

n^2 , $a_1 = S_1 = 1$, $n \geq 2$ 时 $a_n = 2n - 1$, 验证 $n = 1$ 时 $2 \times 1 - 1 = 1 = a_1$, 故通项为 $a_n = 2n - 1$; 若 $S_n = 2^n$, $a_1 = 2$, $n \geq 2$ 时 $a_n = 2^n - 2^{n-1} = 2^{n-1}$, $n = 1$ 时 $2^0 = 1 \neq 2$, 故通项需分段写)。

2. 公差符号判断错误: $d > 0$ 数列递增, $d < 0$ 递减, $d = 0$ 常数列(如 $a_1 = 5$, $d = -2$, 数列是 $\{5, 3, 1, -1, \dots\}$, 递减)。

3. 前 n 项和公式混淆: 已知 a_n 用 $S_n = \frac{n(a_1 + a_n)}{2}$, 已知 d 用 $S_n = na_1 + \frac{n(n-1)}{2}d$, 避免代错参数。

四、等比数列(Geometric Sequence)

4.1 完整定义(Complete Definition)

若一个数列从**第二项**开始，每一项与它前一项的比都等于**同一个非零常数**，则这个数列称为等比数列。

数学表达式:对任意 $n \geq 2$ (n 为正整数), 有 $\frac{a_n}{a_{n-1}} = q$, 其中:

- q 称为“公比”(Common Ratio), **不能为 0**(若 $q = 0$, 则 $a_2 = a_1 \times 0 = 0$, 后续 $a_3 = a_2 \times 0 = 0$, 但 $\frac{a_3}{a_2} = \frac{0}{0}$ 无意义);
- q 可为正或负($q > 0$ 时数列各项同号, $q < 0$ 时数列各项正负交替, 如 $\{2, -4, 8, -16\}$);
- 首项 $a_1 \neq 0$ (若 $a_1 = 0$, 则所有项均为 0, 同样出现 $\frac{0}{0}$ 的无意义情况)。

English Definition:

A sequence is called a geometric sequence if the ratio of each term to its previous term is the **same non-zero constant** starting from the **second term**.

Mathematical expression: For any $n \geq 2$ (n is a positive integer), $\frac{a_n}{a_{n-1}} = q$, where:

- q is called “common ratio”, which **cannot be 0** (if $q = 0$, then $a_2 = a_1 \times 0 = 0$, and subsequent $a_3 = a_2 \times 0 = 0$, but $\frac{a_3}{a_2} = \frac{0}{0}$ is meaningless);
- q can be positive or negative (all terms have the same sign if $q > 0$, and terms alternate between positive and negative if $q < 0$, e.g., $\{2, -4, 8, -16\}$);
- The first term $a_1 \neq 0$ (if $a_1 = 0$, all terms are 0, leading to the meaningless $\frac{0}{0}$).

4.2 核心公式推导(Core Formula Derivation)

(1)通项公式(General Term Formula): $a_n = a_1 q^{n-1}$

推导过程:

由等比数列定义, 有:

$$\frac{a_2}{a_1} = q \rightarrow a_2 = a_1 q;$$

$$\frac{a_3}{a_2} = q \rightarrow a_3 = a_2 q = a_1 q^2;$$

$$\frac{a_4}{a_3} = q \rightarrow a_4 = a_3 q = a_1 q^3;$$

...

以此类推, 第 n 项是首项乘以 q 的 $(n-1)$ 次方, 故:

$$a_n = a_1 q^{n-1}.$$

Derivation Process:

From the definition of geometric sequence:

$$\frac{a_2}{a_1} = q \rightarrow a_2 = a_1 q;$$

$$\frac{a_3}{a_2} = q \rightarrow a_3 = a_2 q = a_1 q^2;$$

$$\frac{a_4}{a_3} = q \rightarrow a_4 = a_3 q = a_1 q^3;$$

...

By analogy, the n -th term is the first term multiplied by q to the power of $(n-1)$, so:

$$a_n = a_1 q^{n-1}.$$

(2)等比中项公式(Geometric Mean Formula):若 a, G, b 成等比数列, 则 $G = \pm\sqrt{ab}$

定义:若三个数 a, G, b 满足 " $\frac{G}{a} = \frac{b}{G}$ "(即 G 与 a 的比等于 b 与 G 的比), 则 G 称为 a 与 b 的等比中项。

推导:

由 $\frac{G}{a} = \frac{b}{G}$, 交叉相乘得 $G^2 = ab$, 故 $G = \pm\sqrt{ab}$;

注意:

- 只有当 $ab > 0$ (即 a 与 b 同号) 时, 等比中项存在;
- 等比中项有两个, 且互为相反数 (如: 2 与 8 的等比中项为 $\pm\sqrt{2 \times 8} = \pm 4$)。

Definition: If three numbers a, G, b satisfy " $\frac{G}{a} = \frac{b}{G}$ " (i.e., the ratio of G to a equals the ratio of b to G), then G is called the geometric mean of a and b .

Derivation:

From $\frac{G}{a} = \frac{b}{G}$, cross-multiply to get $G^2 = ab$, so $G = \pm\sqrt{ab}$;

Notes:

- The geometric mean exists only when $ab > 0$ (i.e., a and b have the same sign);
- There are two geometric means, which are opposites of each other (e.g., the geometric means of 2 and 8 are $\pm\sqrt{2 \times 8} = \pm 4$).

(3)前 n 项和公式(Sum of First n Terms Formula):两种形式($q \neq 1$)

形式 1: $S_n = \frac{a_1(1-q^n)}{1-q}$ (已知 a_1 和 q 时用)

推导方法:错位相减法(Displacement Subtraction):

设 $S_n = a_1 + a_1q + a_1q^2 + \cdots + a_1q^{n-1}$ ①;

两边乘以公比 q , 得 $qS_n = a_1q + a_1q^2 + a_1q^3 + \cdots + a_1q^n$ ②;

① - ② 得: $S_n - qS_n = a_1 - a_1q^n \rightarrow S_n(1-q) = a_1(1-q^n)$;

因 $q \neq 1$ (若 $q = 1$, 分母为 0, 需单独讨论), 故两边除以 $(1-q)$, 得:

$$S_n = \frac{a_1(1-q^n)}{1-q}.$$

Derivation Method: Displacement Subtraction:

Let $S_n = a_1 + a_1q + a_1q^2 + \cdots + a_1q^{n-1}$ ①;

Multiply both sides by the common ratio q : $qS_n = a_1q + a_1q^2 + a_1q^3 + \cdots + a_1q^n$ ②;

Subtract ② from ①: $S_n - qS_n = a_1 - a_1q^n \rightarrow S_n(1 - q) = a_1(1 - q^n)$;

Since $q \neq 1$ (if $q = 1$, the denominator is 0, which needs separate discussion), divide both sides by $(1 - q)$:

$$S_n = \frac{a_1(1-q^n)}{1-q}.$$

形式 2: $S_n = \frac{a_1 - a_nq}{1-q}$ (已知 a_1 、 a_n 和 q 时用)

推导:

将通项公式 $a_n = a_1q^{n-1}$ 变形为 $a_1q^n = a_nq$, 代入形式 1, 得:

$$S_n = \frac{a_1 - a_nq}{1-q}.$$

Derivation:

Transform the general term formula $a_n = a_1q^{n-1}$ into $a_1q^n = a_nq$, substitute into Form 1:

$$S_n = \frac{a_1 - a_nq}{1-q}.$$

特殊情况: 当 $q = 1$ 时

若 $q = 1$, 等比数列为常数列 ($a_1 = a_2 = \cdots = a_n$), 故前 n 项和 $S_n = a_1 + a_1 + \cdots + a_1 = na_1$.

Special Case: When :

If $q = 1$, the geometric sequence is a **constant sequence** ($a_1 = a_2 = \cdots = a_n$), so the sum of the first n terms $S_n = a_1 + a_1 + \cdots + a_1 = na_1$.

4.3 典型例题与解析(Typical Examples & Explanations)

例 1:求等比数列的特定项(Find Specific Term of Geometric Sequence)

题目:在等比数列 $\{a_n\}$ 中, $a_1 = 3$, $q = 2$, 求 a_4 。

英文题目:In the geometric sequence $\{a_n\}$, $a_1 = 3$, $q = 2$, find a_4 .

解析:

1. 确定公式:已知 a_1 和 q , 用通项公式 $a_n = a_1 q^{n-1}$;

2. 代入参数: $n = 4$, $a_1 = 3$, $q = 2$, 得:

$$a_4 = 3 \times 2^{4-1} = 3 \times 2^3 = 3 \times 8 = 24;$$

3. 结论: $a_4 = 24$ 。

Explanation:

1. Determine the formula: Given a_1 and q , use the general term formula

$$a_n = a_1 q^{n-1};$$

2. Substitute parameters: $n = 4$, $a_1 = 3$, $q = 2$, so:

$$a_4 = 3 \times 2^{4-1} = 3 \times 2^3 = 3 \times 8 = 24;$$

3. Conclusion: $a_4 = 24$ 。

例 2:已知前 n 项和求公比和特定项(Find q and Specific Term Given Sum of First n Terms)

题目:在等比数列 $\{a_n\}$ 中, $a_1 = 2$, $S_3 = 14$, 求 q 和 a_3 。

英文题目:In the geometric sequence $\{a_n\}$, $a_1 = 2$, $S_3 = 14$, find q and a_3 .

解析:

1. 讨论 q 的情况:

• 若 $q = 1$, 则 $S_3 = 3a_1 = 3 \times 2 = 6 \neq 14$, 故 $q \neq 1$, 用前 n 项和公式 $S_n =$

$$\frac{a_1(1-q^n)}{1-q};$$

1. 列方程求解 q :

代入 $a_1 = 2$, $n = 3$, $S_3 = 14$, 得:

$$14 = \frac{2(1-q^3)}{1-q} \rightarrow \text{两边除以 } 2: 7 = \frac{1-q^3}{1-q};$$

由立方差公式 $1 - q^3 = (1 - q)(1 + q + q^2)$ ($q \neq 1$ 时), 代入得:

$$7 = 1 + q + q^2 \rightarrow \text{整理为一元二次方程: } q^2 + q - 6 = 0;$$

$$\text{因式分解: } (q + 3)(q - 2) = 0 \rightarrow \text{解得 } q = -3 \text{ 或 } q = 2;$$

2. 求 a_3 :

- 当 $q = -3$ 时, $a_3 = a_1 q^2 = 2 \times (-3)^2 = 2 \times 9 = 18$;
- 当 $q = 2$ 时, $a_3 = a_1 q^2 = 2 \times 2^2 = 2 \times 4 = 8$;

1. 结论: $q = -3$ 且 $a_3 = 18$, 或 $q = 2$ 且 $a_3 = 8$ 。

Explanation:

1. Discuss the case of q :

- If $q = 1$, then $S_3 = 3a_1 = 3 \times 2 = 6 \neq 14$, so $q \neq 1$, use the sum formula

$$S_n = \frac{a_1(1-q^n)}{1-q};$$

1. List the equation to solve q :

Substitute $a_1 = 2$, $n = 3$, $S_3 = 14$:

$$14 = \frac{2(1-q^3)}{1-q} \rightarrow \text{Divide both sides by } 2: 7 = \frac{1-q^3}{1-q};$$

Use the difference of cubes formula $1 - q^3 = (1 - q)(1 + q + q^2)$ (when $q \neq 1$), substitute:

$$7 = 1 + q + q^2 \rightarrow \text{Rearrange into a quadratic equation: } q^2 + q - 6 = 0;$$

$$\text{Factorize: } (q + 3)(q - 2) = 0 \rightarrow \text{Solve for } q = -3 \text{ or } q = 2;$$

2. Find a_3 :

- When $q = -3$, $a_3 = a_1 q^2 = 2 \times (-3)^2 = 2 \times 9 = 18$;
- When $q = 2$, $a_3 = a_1 q^2 = 2 \times 2^2 = 2 \times 4 = 8$;

1. Conclusion: $q = -3$ and $a_3 = 18$, or $q = 2$ and $a_3 = 8$.

4.4 常见易错点(Common Pitfalls)

1. 忽略 $q \neq 0$ 和 $a_1 \neq 0$:如“数列 $\{0,0,0\}$ ”不是等比数列(因 $\frac{0}{0}$ 无意义);
2. 忘记讨论 $q = 1$:用前 n 项和公式前必须先判断 q 是否为 1($q = 1$ 时公式为 $S_n = na_1$, $q \neq 1$ 时用错位相减推导的公式);
3. 等比中项符号错误:若 a 与 b 同号, 等比中项有两个(正负); 若异号, 无等比中项(如 -2 和 8 异号, $\sqrt{(-2) \times 8} = \sqrt{-16}$ 无实数意义)。

五、等差与等比数列对比(Comparison Between Arithmetic and Geometric Sequences)

对比维度	等差数列 (Arithmetic Sequence)	等比数列 (Geometric Sequence)
定义核心	差为常数($a_n - a_{n-1} = d$)	比为常数($\frac{a_n}{a_{n-1}} = q \neq 0$)
通项公式	$a_n = a_1 + (n - 1)d$	$a_n = a_1 q^{n-1} (a_1 \neq 0)$
中项公式	$A = \frac{a+b}{2}$ (1 个)	$G = \pm\sqrt{ab}$ (2 个, $ab > 0$)
前 n 项和	$S_n = \frac{n(a_1+a_n)}{2}$ 或 $na_1 + \frac{n(n-1)}{2}d$	$S_n = \begin{cases} na_1 & (q = 1) \\ \frac{a_1(1 - q^n)}{1 - q} & (q \neq 1) \end{cases}$
单调性	$d > 0$ 递增, $d < 0$ 递减, $d = 0$ 常数列	$a_1 > 0$ 时: $q > 1$ 递增, $0 < q < 1$ 递减; $a_1 < 0$ 时相反
项的符号	全同号或正负交替($d \neq 0$ 时)	$q > 0$ 全同号, $q < 0$ 正负交替